

MEDIUM LEVEL QUESTIONS

NAME: B. HARSHVARDHAN
REG NO.: 192425201

1. Palindrome Number

The screenshot shows the SIMATS Online Programming Lab interface. On the left, a sidebar lists seven questions, with 'Q1 Palindrome number/string' selected. The main area displays the question title 'Q1 Palindrome number/string' with a 10-mark value. Below the title is a text input field containing '151'. To the right of the input field is an 'Output' box showing '151 is a Palindrome'. The 'Your Code' section contains a Java program that checks if the input string is a palindrome by reversing it and comparing the two strings. The code is as follows:

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         String input = sc.nextLine().trim();
7         String reversed = new StringBuilder(input).reverse().toString();
8
9         if (input.equals(reversed)) {
10             System.out.println(input + " is a Palindrome");
11         } else {
12             System.out.println(input + " is not a Palindrome");
13         }
14     }
15 }
```

At the bottom of the code editor are 'Run' and 'Save' buttons. The bottom status bar indicates '7/7 answered'.

2. Perfect Number

The screenshot shows the SIMATS Online Programming Lab interface. On the left, a sidebar lists seven questions, with 'Q2 First n perfect numbers' selected. The main area displays the question title 'Q2 First n perfect numbers' with a 10-mark value. Below the title is a text input field containing '28'. To the right of the input field is an 'Output' box showing 'Enter a number: Perfect Number'. The 'Your Code' section contains a Java program that checks if the input number is a perfect number by summing its divisors and comparing the sum to the input number. The code is as follows:

```
1 import java.util.Scanner;
2
3 class PerfectNumber {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         System.out.print("Enter a number: ");
7         int n = sc.nextInt();
8         int sum = 0;
9         for (int i = 1; i < n; i++) {
10             if (n % i == 0) {
11                 sum = sum + i;
12             }
13         }
14         if (sum == n) {
15             System.out.println("Perfect Number");
16         } else {
17             System.out.println("Not a Perfect Number");
18         }
19     }
20 }
```

At the bottom of the code editor are 'Run' and 'Save' buttons. The bottom status bar indicates '7/7 answered'.

3. Decimal to Binary

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192425201RD

QUESTIONS

Q1
Palindrome number/string
10 marks

Q2
First n perfect numbers
10 marks

Q3
Decimal number equivalent to binary number and octal number
10 marks

Q4
Nth max number and nth min number in array then sum of it and difference of it
10 marks

Q5
Find the average positive and negative numbers upto -1
10 marks

Q6
Removing duplicates in array
10 marks

Q7
Matrix multiplication
10 marks

7/7 answered

Q3
Decimal number equivalent to binary number and octal number
10 marks

Decimal number equivalent to binary number and octal number

Your Code

```
1 import java.util.Scanner;
2
3 class BinaryToDecimal {
4     public static void main(String[] args) {
5
6         Scanner sc = new Scanner(System.in);
7
8         System.out.print("Enter Binary Number: ");
9         int binary = sc.nextInt();
10
11         int decimal = 0, power = 0;
12
13         while (binary > 0) {
14             int digit = binary % 10;
15             decimal = decimal + digit * (int)Math.pow(2, power);
16             power++;
17             binary = binary / 10;
18         }
19
20         System.out.println("Decimal = " + decimal);
21     }
22 }
```

Run Save

Input

1010

Output

Enter Binary Number: Decimal = 10

4. Nth Maximum and Nth Minimum Number in an Array, Then Find Their Sum and Difference.

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QUESTIONS

Q1
Palindrome number/string
10 marks

Q2
First n perfect numbers
10 marks

Q3
Decimal number equivalent to binary number and octal number
10 marks

Q4
Nth max number and nth min number in array then sum of it and difference of it
10 marks

Q5
Find the average positive and negative numbers upto -1
10 marks

Q6
Removing duplicates in array
10 marks

Q7
Matrix multiplication
10 marks

7/7 answered

Q4
Nth max number and nth min number in array then sum of it and difference of it
10 marks

Nth max number and nth min number in array then sum of it and difference of it

Your Code

```
1 import java.util.Arrays;
2
3 public class NthMaxMin {
4     public static void main(String[] args) {
5
6         int[] a = {10, 40, 20, 50, 30};
7         int n = 2;
8
9         Arrays.sort(a);
10        int min = a[n - 1];
11        int max = a[a.length - n];
12
13        System.out.println("Nth Minimum = " + min);
14        System.out.println("Nth Maximum = " + max);
15        System.out.println("Sum = " + (min + max));
16        System.out.println("Difference = " + (max - min));
17    }
18 }
```

Run Save

Input

stdin input

Output

Nth Minimum = 20
Nth Maximum = 40
Sum = 60
Difference = 20

5. Find the Average of Positive and Negative Numbers (Until -1 is Entered)

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QUESTIONS

Q1
Palindrome number/string
10 marks

Q2
First n perfect numbers
10 marks

Q3
Decimal number equivalent to binary number and octal number
10 marks

Q4
Nth max number and nth min number in array then sum of it and difference of it
10 marks

Q5
Find the average positive and negative numbers upto -1
10 marks

Q6
Removing duplicates in array
10 marks

Q7
Matrix multiplication
10 marks

7/7 answered

Q5 Find the average positive and negative numbers upto -1 10 marks

Find the average positive and negative numbers upto -1

Your Code

```
1 import java.util.Scanner;
2
3 class Average {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         int num;
7         int psum = 0, nsum = 0;
8         int pcount = 0, ncount = 0;
9         System.out.println("Enter numbers (-1 to stop):");
10
11         while (true) {
12             num = sc.nextInt();
13
14             if (num == -1)
15                 break;
16
17             if (num > 0) {
18                 psum += num;
19                 pcount++;
20             } else {
21                 nsum += num;
22                 ncount++;
23             }
24         }
25
26         if (pcount > 0)
27             System.out.println("Positive Average = " + (double) psum /
28
29         if (ncount > 0)
30             System.out.println("Negative Average = " + (double) nsum /
31     }
```

Input

-8
4
-12
-1

Output

Enter numbers (-1 to stop):
Positive Average =
8.533333333333334
Negative Average = -10.0

6. Removing Duplicates in an array.

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QUESTIONS

Q1
Palindrome number/string
10 marks

Q2
First n perfect numbers
10 marks

Q3
Decimal number equivalent to binary number and octal number
10 marks

Q4
Nth max number and nth min number in array then sum of it and difference of it
10 marks

Q5
Find the average positive and negative numbers upto -1
10 marks

Q6
Removing duplicates in array
10 marks

Q7
Matrix multiplication
10 marks

7/7 answered

Q6 Removing duplicates in array 10 marks

Removing duplicates in array

Your Code

```
1 import java.util.Scanner;
2 class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         int arr[] = new int[n];
7         for (int i = 0; i < n; i++) {
8             arr[i] = sc.nextInt();
9         }
10         for (int i = 0; i < n; i++) {
11             for (int j = i + 1; j < n; j++) {
12                 if (arr[i] == arr[j]) {
13                     for (int k = j; k < n - 1; k++) {
14                         arr[k] = arr[k + 1];
15                     }
16                     n--;
17                     j--;
18                 }
19             }
20         }
21         System.out.println("Array without duplicates:");
22         for (int i = 0; i < n; i++) {
23             System.out.print(arr[i] + " ");
24         }
25         sc.close();
26     }
27 }
```

Input

5
1 4 8 1 5

Output

Array without duplicates:
1 4 8 5

 

7. Matrix Multiplication

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 **BADHRI HARSHA VARDHAN**
1924252019D

QUESTIONS

Q1
Palindrome number/string
10 marks

Q2
First n perfect numbers
10 marks

Q3
Decimal number equivalent to
binary number and octal
number
10 marks

Q4
nth max number and nth min
number in array then sum of it
and difference of it
10 marks

Q5
Find the average positive and
negative numbers upto -1
10 marks

Q6
Removing duplicates in array
10 marks

Q7
Matrix multiplication
10 marks

7/7 answered

Q7 Matrix multiplication 10 marks

Matrix multiplication

Your Code

```
1 class MatrixMultiplication {
2     public static void main(String[] args){
3         int a[][] = {{1,2},{3,4}};
4         int b[][] = {{5,6},{7,8}};
5         int c[][] = new int[2][2];
6         for (int i=0;i<2;i++){
7             for (int j=0;j<2;j++){
8                 for (int k=0;k<2;k++){
9                     c[i][j] += a[i][k]*b[k][j];
10                }
11            }
12        }
13        System.out.println("Result Matrix:");
14        for (int i=0;i<2;i++){
15            for (int j=0;j<2;j++){
16                System.out.print(c[i][j]+" ");
17            }
18            System.out.println();
19        }
20    }
21 }
22
23
```

Run Save

Input

Output

Result Matrix:
19 22
43 50